数值分析第二十次作业

崔正平 2400010714

1. 证明对于任意参数 α , 下列 Runge-Kutta 格式是二阶的:

$$\begin{cases} y_{n+1} = y_n + \dfrac{h}{2}(K_2 + K_3), \\ K_1 = f(t_n, y_n), \\ K_2 = f(t_n + \alpha h, y_n + \alpha h K_1), \\ K_3 = f(t_n + (1-\alpha)h, y_n + (1-\alpha)hK_2). \end{cases}$$

解答.

记 $f(t_n, y_n) = f_n$, 由 Taylor 展开可得截断误差

$$\begin{aligned} & y(t_{n+1}) - y(t_n) - \frac{h}{2}(K_2 + K_3) \\ = & hf_n + \frac{h^2}{2}(\partial_t f_n + \partial_y f_n \cdot f_n) \\ & - \frac{h}{2}(f_n + \alpha h \partial_t f_n + \alpha h \partial_y f_n \cdot f_n + f_n + (1-\alpha)h\partial_t f_n + (1-\alpha)h\partial_y f_n \cdot f_n) + \mathcal{O}(h^3) \\ = & \mathcal{O}(h^3), \end{aligned}$$

故该 Runge-Kutta 格式是二阶的. □

2. 对具有 Butcher 表 $(\boldsymbol{A}, \boldsymbol{b}, \boldsymbol{c})$ 的 s 级 Runge-Kutta 方法, 证明其放大因子为

$$R(z) = 1 + z\boldsymbol{b}^{\mathrm{T}}(\boldsymbol{I} - z\boldsymbol{A})^{-1} \cdot \boldsymbol{1} = \frac{\det(\boldsymbol{I} - z\boldsymbol{A} + z\boldsymbol{1} \cdot \boldsymbol{b}^{\mathrm{T}})}{\det(\boldsymbol{I} - z\boldsymbol{A})}.$$

解答.

由降阶公式可得

$$\begin{aligned} \frac{\det(\boldsymbol{I} - z\boldsymbol{A} + z\boldsymbol{1} \cdot \boldsymbol{b}^{\mathrm{T}})}{\det(\boldsymbol{I} - z\boldsymbol{A})} &= \det((\boldsymbol{I} - z\boldsymbol{A})^{-1}(\boldsymbol{I} - z\boldsymbol{A} + z\boldsymbol{1} \cdot \boldsymbol{b}^{\mathrm{T}})) \\ &= \det(\boldsymbol{I} + z(\boldsymbol{I} - z\boldsymbol{A})^{-1}\boldsymbol{1} \cdot \boldsymbol{b}^{\mathrm{T}}) \\ &= 1 + z\boldsymbol{b}^{\mathrm{T}}(\boldsymbol{I} - z\boldsymbol{A})^{-1} \cdot \boldsymbol{1}. \end{aligned}$$

□